# CASE STUDY ON THE VOLKSWAGEN EMISSIONS SCANDAL (2015)

*Engineering Ethics Case Study – Defeat Devices*

## Technical Overview

The Volkswagen emissions scandal involved software known as a "defeat device" installed in certain diesel vehicles. The software could recognize conditions associated with laboratory emissions testing and change engine operation to reduce nitrogen oxide (NOx) emissions during the test. Under many normal road-driving conditions, the vehicles could produce substantially higher NOx emissions.

## When – How – Why?

The affected vehicles were developed and sold before 2015. In 2014, researchers from West Virginia University, working with the International Council on Clean Transportation (ICCT), conducted on-road testing and identified a major difference between laboratory and real-world emissions. In September 2015, the U.S. Environmental Protection Agency (EPA) issued a Notice of Violation, after which Volkswagen acknowledged the use of software capable of manipulating emissions behaviour during testing.

| **Condition** | **Software Behaviour** | **Result** |
|---|---|---|
| Laboratory test | Activates lower-emission strategy | Lower measured NOx |
| Normal driving | Uses a different operating strategy | Higher NOx |

The failure was not simply an accidental software bug. The system was capable of deliberately changing its behaviour when it detected test-like conditions.

- **Performance requirements:** Engineers had to balance vehicle performance with strict emissions limits.
- **Software workaround:** Software was used to achieve favourable test results instead of consistent real-world compliance.
- **Testing limitations:** Controlled laboratory tests did not fully represent actual road conditions.
- **Weak verification:** Compliance-critical software did not receive sufficient independent scrutiny.
- **Organizational pressure:** Performance, cost, schedule, and product expectations could encourage unacceptable compromises.

A systems-thinking approach focuses on why the engineering and organizational system allowed the failure rather than focusing only on individual responsibility. The key issue is why there were not strong enough barriers to prevent or detect test-specific behaviour.

| System Area | Weakness | Effect |
|---|---|---|
| Requirements | Compliance could become a test-passing objective | Encouraged optimization for testing |
| Software | System could detect test-like conditions | Allowed different test behaviour |
| Testing | Reliance on laboratory conditions | Real-world discrepancy was harder to detect |
| Verification | Insufficient independent review | Problematic logic could remain undetected |
| Governance | Business and engineering pressures | Ethical concerns could be overlooked |

## Engineering Ethics Perspective

Writing software specifically to make a product appear compliant during a regulatory test while allowing different real-world behaviour is ethically unacceptable. Engineering responsibility extends to customers, regulators, society, and the environment. The case shows that technical capability must be combined with ethical judgment and effective system-level controls.

## Architectural Changes

- Separate and protect compliance-critical emissions-control functions.
- Prevent software from selecting a different emissions strategy merely because a test is detected.
- Maintain secure audit logs of important emissions-control decisions and software changes.
- Enforce critical emissions constraints through protected system layers.

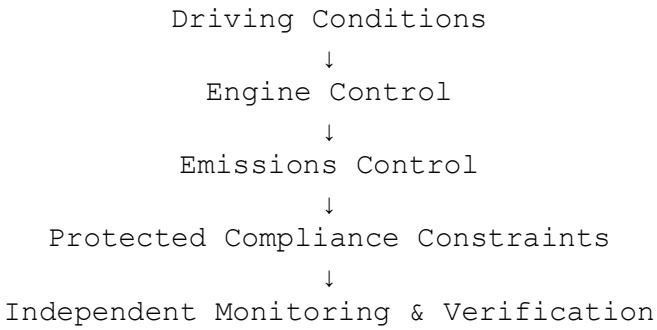
## Testing Changes

- Continue laboratory testing for controlled measurements.
- Use independent testing teams for verification.
- Include real-world on-road emissions testing.
- Use randomized and unpredictable testing conditions.
- Repeat emissions tests after major software or calibration changes.

## Process Changes

- Require peer and independent review of emissions-related software.
- Use version control and traceability for compliance-critical changes.
- Flag software that detects regulatory test conditions.
- Require documented justification and independent approval before release.
- Include ethics and compliance review in the development process.

### Redesigned System Principle




The redesigned system must maintain emissions compliance during real-world operation, not only during testing. Multiple independent controls should make intentional test manipulation difficult to implement and easy to detect.

## Conclusion

The Volkswagen emissions scandal demonstrates that engineering failures can result from software design, testing limitations, verification weaknesses, and organizational pressures. The central lesson is that compliance must represent the actual purpose of a requirement rather than simply producing a passing test result.

A reliable solution requires protected compliance architecture, independent and real-world testing, software auditability, strong verification, and an organizational culture that places genuine compliance and public responsibility above short-term performance targets.